Commure Data Operations Analyst — Case Study
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How I approached the case

My first step was to learn why work was getting stuck and then I could propose a solution. There can be many similarities between a missing record, an unsuccessful submission and an unpaid claim; but the actions are different.

The encounter analysis was performed with the data provided with the dataset, and the results were checked. In the operational problems I have suggested the problems, actions and any assumptions have been clearly written. My hope is to make processing faster as to not duplicate recording and to maintain accuracy of recording.

1. Workers' compensation claims

Team is finishing 50 Claims out of 500 per day. That is 10 percent of the 500 claims that were initially filed each day.

First, I would take a look at a few examples of blocked claims with successful claims. I would trace each claim from the collection of the documents to the PDF creation to the internal tool to the mailing vendor. This would indicate if the major issue is the tool, the working process or an external dependency.

If the tool is failing, then I would provide the engineer a repeatable example of the most common failure. If I had just 2 engineering hours a week, I would spend my time on one change that is going to take a lot of work out of my job.

I would observe operators performing the process and adjust the instructions, if needed, by ensuring a checklist was used on a small batch of operators. I would also look for missing documents, wrong mailing information and vendor requirements.

Once the changes are made, I would be able to keep track of what packets were successful, how long they took to process, and repeat errors, as well as packet accuracy and the oldest unfinished claims. A PDF upload would be considered a non-completed mailing if no vendor confirmation.

The aim is set to be over 500 confirmed mailings per day, meaning that the backlog can start to decrease. I would test that target using a quick capacity test.

2. Missing closed encounters

I first checked how the two files were structured. The EHR contains separate procedure rows, while the database contains procedure lists. Comparing their raw row counts would therefore give the wrong picture.

I cleaned the patient and provider names, standardized the dates and compared distinct encounters using:

Patient name + date of service + rendering provider

The results were:

Measure	Result
Valid EHR encounters	6,612
Distinct database encounters	6,374
Matched encounters	6,330
Unmatched EHR encounters	282
Match rate	95.74%
Database-only encounters	44

I then divided the 282 unmatched encounters into groups that could guide the investigation:

- **205 marker-associated records:** These contain markers such as `NORCM`. Their meanings and exclusion rules need confirmation.
- **8 possible provider discrepancies:** A database-only record has the same patient and date but a different provider. The ordinary procedure-code sets also agree.
- **69 remaining unexplained records:** These need further checking against extraction and import logs.

The markers are a good hint, but can not be used to determine the loss of a record. Successfully imported three encounters with listed markers.

In unmarked games, Liam Young had 34 unmatched records from 622 games which works out to 5.47%. That subset also had the highest monthly unmatched rate in September. These patterns are used to determine where to begin investigation, but do not determine the cause. Also identified two incomplete EHR rows and 108 database rows outside of the distinct encounter count. Prior to making operational corrections, I would pursue investigation of these separately. I built a Retool dashboard to show the results and a searchable review table. The complete encounter list is in `missing_review_queue.csv`. I would not describe all 282 records as lost revenue because the files do not contain enough financial information to support that claim.

3. Resolving “claim not found”

For 10,000 claims from 150 insurers, I'd not choose to begin with individual phone calls. I would first break the claims down by Payer, by Submission Batch, by Age and by the last stage of processing which I could confirm.

Claims that are as close as possible to the deadline would be filed first. Of this set, I think the amount that I would collect and the amount of work that I would have to do to fix the problem.

I would confirm if each claim was never sent, was rejected, was accepted and searched incorrectly or if they were accepted, but no internal status update made. Transmission acknowledgments would aid in tracking the submission but not as proof of payment.

The action would be based on evidence, reject incorrect information, track missing submissions, enhance information searching by the payer or update the payer record. I would make sure to see if there was a claim before I resubmitted.

If 10 operators can complete 25 of their investigations that have been verified per day, then 10 operators can complete 250 investigations per day. This means 50 new exceptions will arrive each day, so the actual reduction will be 200, which equates to an estimated 50 working days. These are assumptions I would want to change to measured results after a pilot.

4. Understanding the payment differences

This CPT code is for Medicare payments for 97110.

Locality codes are for state of Alaska and Mississippi. Medicare compensates for variations in cost of practice in different areas.The basic calculation is a combination of work, practice-expense and malpractice values, followed by a geographic adjustment to each of these values, and then multiplying the total number by a conversion factor.

Using the supplied figures:

$38.73 − $27.55 = $11.18

The Alaska amount is approximately **40.58% higher** than the Mississippi amount.

To reproduce the exact prices, I would need the service date, fee-schedule version, setting and applicable adjustments. I would verify these rather than choose values that simply produce the expected answer.

B. Aetna payments for 97140

My first guess to look into would be multiple procedure payment reduction for the same day services.

Aetna has published a therapy policy for reducing the practice-expense portion of follow-up services by 50%. It does not take the half to reduce the entire payment. There is a need to check the applicable contract/service-date policy. A $40 allowance is made up of $24 in practice

expense and $16 in other expenses. If these other expenses are $16, then the allowance would be reduced by:

($24 × 50%) + $16 = $28

Therefore, I would not assume this rule alone fully explains the approximate $20 - versus -40$ payouts. I would compare units, allowed amounts, patient responsibility and the remittance adjustments before confirming the reason.

C. Correcting the therapy units

Assuming the documented minutes qualify, the timed treatment total is:

38 + 24 = 62 minutes

This supports four timed units. After allocating the full 15-minute blocks, nine minutes remain for 97112 and eight for 97110. The fourth timed unit therefore goes to 97112.G0283 is untimed. For the single service described, it should be one unit, and its minutes should be excluded from the timed calculation.

Code	Supplied units	Corrected units
97110	2	2
97112	1	2
G0283	3	1

The goal is to collect the correct payment for documented care, rather than increase units unnecessarily.

Putting the plan into practice

I think I'd start with 8 India operators for mailing, 2 to review encounters and 10 claim tracers, with 1 Mountain View operator for each of them.I would allocate 60 minutes of the engineer's time for mailings, 30 minutes for problem solving and 30 minutes for claim chasing. Following the pilot there would be some further adjustments to these assignments in terms of workload, quality and results.